UNIVERSITY OF SCIENCE AND TECHNOLOGY
OF SOUTHERN PHILIPPINES
Alubijid | Balubal |Cagayan de Oro | Claveria | Jasaan | Oroquieta | Panaon | Villanueva

# Lab Activity 2

IT325 Machine Learning | Performing EDA for the PIT Dataset

| | |
|---|---|
| **Type** | Laboratory Activity |
| **Topic Focus** | Exploratory data analysis for text classification datasets |
| **Time Allotment** | 1 laboratory meeting |
| **PIT Connection** | Students generate EDA outputs that will support preprocessing and model selection |
| **Mode** | Individual hands on work |

## Learning Outcomes

1. Load and inspect a text classification dataset using Python.
2. Detects missing values, duplicates, class imbalance, and basic text characteristics.
3. Present and interpret simple EDA visualizations.

## Task Overview

Students will implement the EDA plan prepared in Lecture Activity 1. The output of this task will become part of the PIT notebook and may also be used in the final presentation.

## Laboratory Tasks

1. Load the dataset in pandas and display the first few rows.
2. Show the dataset shape, column names, and data types.
3. Check for missing values and duplicate rows.
4. Show the distribution of the target variable.
5. Create at least five  EDA visualizations relevant to the dataset.
7. In a Text Cell Write three to five observations based on the analysis.

## Minimum Required Outputs

1. A Jupyter Notebook or Colab notebook with complete code
2. At least two visuals such as class distribution or text length distribution
3. A short interpretation paragraph written in clear academic language

## Submission Reminder

The notebook should be organized and readable. Use meaningful section titles such as dataset overview, data quality checks, class distribution, text characteristics, and interpretation.

## Simple Scoring Guide

Use the checklist below to evaluate the submitted work for this activity.

| Criteria | Description | Points |
|---|---|---|
| **Correct Execution** | Notebook runs and shows the required EDA steps. | 10 |
| **Visual Outputs** | At least two relevant visuals are included. | 10 |
| **Interpretation** | Observations are clear and connected to the dataset. | 5 |

# Exploratory Data Analysis: Student Depression Dataset

## Lab Activity – Text Classification EDA

John Lemar L. Gonzales - BSIT3-R15

---

## 1. Setup and Imports

[ ]:
```
# import the tools
import pandas as pd              # for working with tables/data
import numpy as np               # for math stuff
import matplotlib.pyplot as plt  # for making charts
import seaborn as sns            # for nicer-looking charts
import warnings
warnings.filterwarnings('ignore') # hide unnecessary warnings

# This just makes our charts look cleaner
sns.set_theme(style="whitegrid")
plt.rcParams["figure.dpi"] = 120

print("All libraries loaded!")
```

## 2. Dataset Overview

```
df = pd.read_csv(r"C:\Lemar\GithubRepositories\student-depression-detector\notebooks\Student Depression Dataset.csv")

print("First 5 rows of the dataset:")
df.head()
```

```
First 5 rows of the dataset:
```

| | id | Gender | Age | City | Profession | Academic Pressure | Work Pressure | CGPA | Study Satisfaction | Job Satisfaction | Sleep Duration | Dietary Habits | Degree | Have you ever had suicidal thoughts ? | Work/Study Hours | Financial Stress |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **0** | 2 | Male | 33.0 | Visakhapatnam | Student | 5.0 | 0.0 | 8.97 | 2.0 | 0.0 | 5-6 hours | Healthy | B.Pharm | Yes | 3.0 | 1.0 |
| **1** | 8 | Female | 24.0 | Bangalore | Student | 2.0 | 0.0 | 5.90 | 5.0 | 0.0 | 5-6 hours | Moderate | BSc | No | 3.0 | 2.0 |
| **2** | 26 | Male | 31.0 | Srinagar | Student | 3.0 | 0.0 | 7.03 | 5.0 | 0.0 | Less than 5 hours | Healthy | BA | No | 9.0 | 1.0 |
| **3** | 30 | Female | 28.0 | Varanasi | Student | 3.0 | 0.0 | 5.59 | 2.0 | 0.0 | 7-8 hours | Moderate | BCA | Yes | 4.0 | 5.0 |
| **4** | 32 | Female | 25.0 | Jaipur | Student | 4.0 | 0.0 | 8.13 | 3.0 | 0.0 | 5-6 hours | Moderate | M.Tech | Yes | 1.0 | 1.0 |

```python
print("Shape (rows, columns):", df.shape)

print("\nColumn Names:")
for col in df.columns:
    print(" -", col)
```

```
Shape (rows, columns): (27901, 18)

Column Names:
 - id
 - Gender
 - Age
 - City
 - Profession
 - Academic Pressure
 - Work Pressure
 - CGPA
 - Study Satisfaction
 - Job Satisfaction
 - Sleep Duration
 - Dietary Habits
 - Degree
 - Have you ever had suicidal thoughts ?
 - Work/Study Hours
 - Financial Stress
 - Family History of Mental Illness
 - Depression
```

```python
# Check data type
print("Data Types of Each Column:")
print(df.dtypes)
```

```
Data Types of Each Column:
id                                       int64
Gender                                     str
Age                                    float64
City                                       str
Profession                                 str
Academic Pressure                      float64
Work Pressure                          float64
CGPA                                   float64
Study Satisfaction                     float64
Job Satisfaction                       float64
Sleep Duration                             str
Dietary Habits                             str
Degree                                     str
Have you ever had suicidal thoughts ?      str
Work/Study Hours                       float64
Financial Stress                       float64
Family History of Mental Illness           str
Depression                               int64
dtype: object
```

## 3. Data Quality Checks

```python
# Check for missing (empty) values in each column
print("Missing values per column:")
print(df.isnull().sum())
```

```
Missing values per column:
id                                       0
Gender                                   0
Age                                      0
City                                     0
Profession                               0
Academic Pressure                        0
Work Pressure                            0
CGPA                                     0
Study Satisfaction                       0
Job Satisfaction                         0
Sleep Duration                           0
Dietary Habits                           0
Degree                                   0
Have you ever had suicidal thoughts ?    0
Work/Study Hours                         0
Financial Stress                         3
Family History of Mental Illness         0
Depression                               0
dtype: int64
```

```python
# Check for duplicate rows
duplicates = df.duplicated().sum()
print("Number of duplicate rows:", duplicates)
```

```
Number of duplicate rows: 0
```

## 4. Class Distribution (Target Variable)

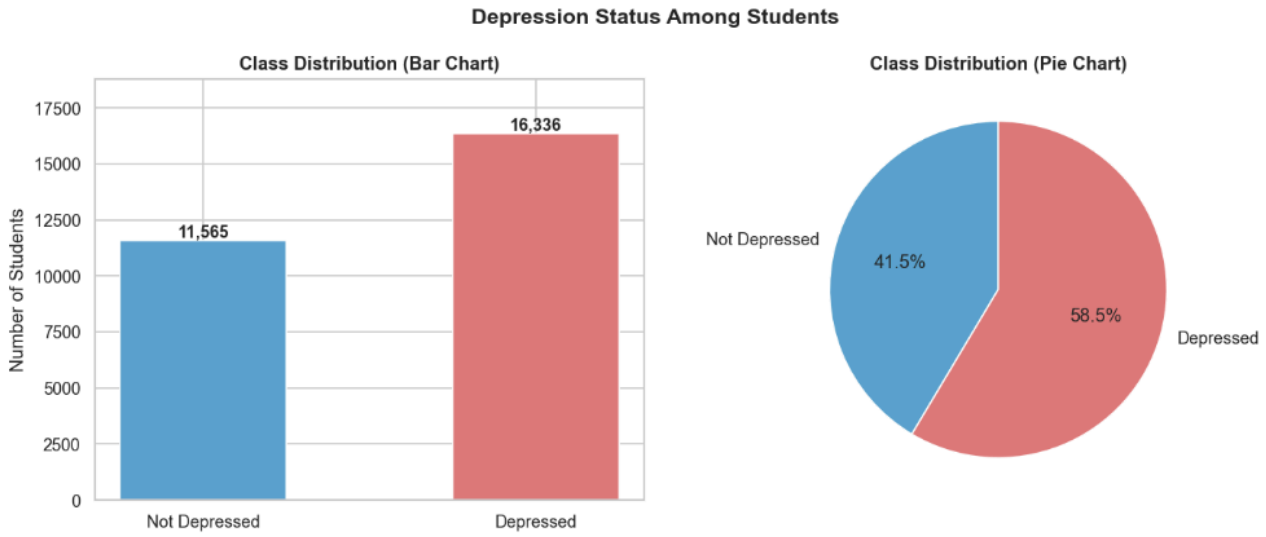
```python
# Count how many students are depressed (1) vs not depressed (0)
print("Depression column value counts:")
print(df["Depression"].value_counts())

print()

# Show percentage also
total = len(df)
for val, count in df["Depression"].value_counts().items():
    label = "Depressed" if val == 1 else "Not Depressed"
    print(f"{label}: {count} students ({count/total*100:.1f}%)")
```

```
Depression column value counts:
Depression
1    16336
0    11565
Name: count, dtype: int64

Depressed: 16336 students (58.5%)
Not Depressed: 11565 students (41.5%)
```

## 5. EDA Visualizations

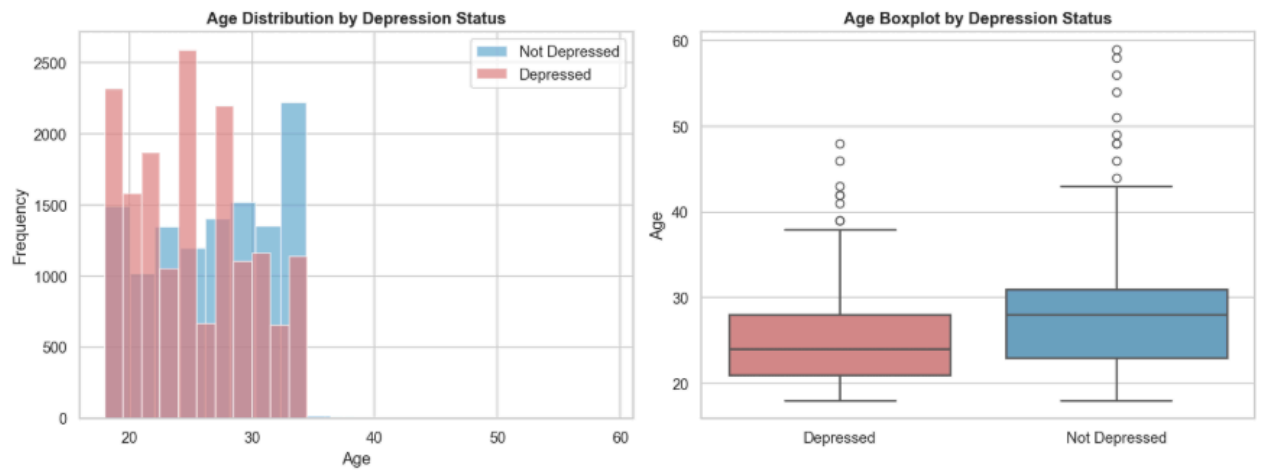
### 5.1 Target Class Distribution

```python
# We want to see how many students are depressed vs not depressed
labels = ["Not Depressed", "Depressed"]
counts = df["Depression"].value_counts().sort_index().values
colors = ["#5BA4CF", "#E07B7B"]

# Create two charts side by side
fig, axes = plt.subplots(1, 2, figsize=(12, 5))

# --- Left chart: Bar chart ---
bars = axes[0].bar(labels, counts, color=colors, edgecolor="white", width=0.5)
axes[0].set_title("Class Distribution (Bar Chart)", fontweight="bold")
axes[0].set_ylabel("Number of Students")
axes[0].set_ylim(0, max(counts) * 1.15)

# Add the number on top of each bar
for bar, count in zip(bars, counts):
    axes[0].text(bar.get_x() + bar.get_width()/2,
                 bar.get_height() + 150,
                 f"{count:,}", ha="center", fontsize=11, fontweight="bold")

# --- Right chart: Pie chart ---
axes[1].pie(counts, labels=labels, autopct="%1.1f%%",
            colors=colors, startangle=90)
axes[1].set_title("Class Distribution (Pie Chart)", fontweight="bold")

plt.suptitle("Depression Status Among Students", fontsize=14, fontweight="bold")
plt.tight_layout()
plt.savefig("viz1_class_distribution.png", bbox_inches="tight")
plt.show()
```



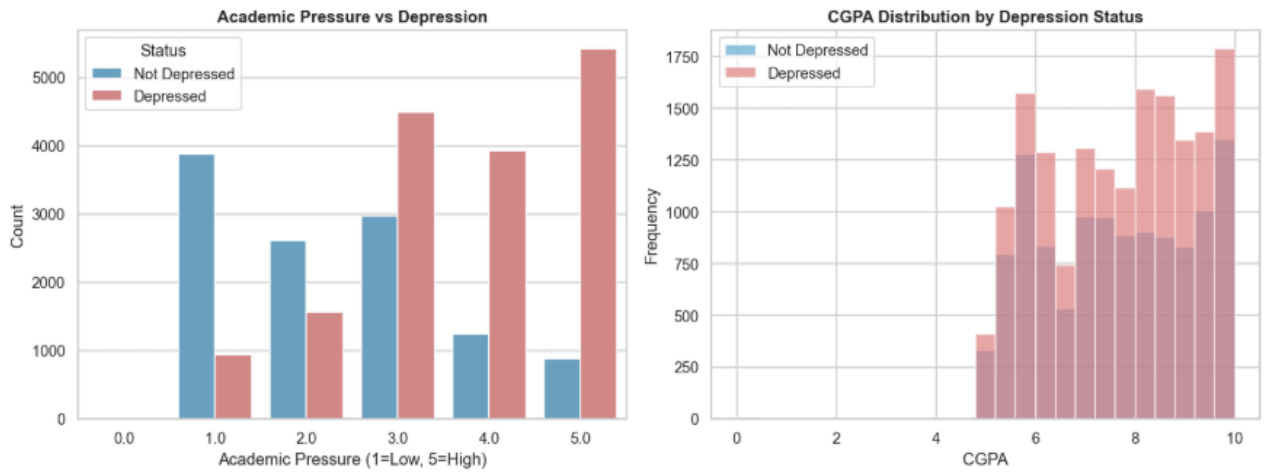

## 5.2 Age Distribution by Depression Status

```
# Add a label column so charts are easier to read
df["Depression_Label"] = df["Depression"].map({0: "Not Depressed", 1: "Depressed"})

fig, axes = plt.subplots(1, 2, figsize=(13, 5))

# --- Left: Histogram to see age spread ---
for val, label, color in [(0, "Not Depressed", "#5BA4CF"), (1, "Depressed", "#E07B7B")]:
    subset = df[df["Depression"] == val]["Age"]
    axes[0].hist(subset, bins=20, alpha=0.65, label=label, color=color, edgecolor="white")

axes[0].set_title("Age Distribution by Depression Status", fontweight="bold")
axes[0].set_xlabel("Age")
axes[0].set_ylabel("Frequency")
axes[0].legend()

# --- Right: Boxplot to compare age ranges ---
sns.boxplot(data=df, x="Depression_Label", y="Age",
            palette={"Not Depressed": "#5BA4CF", "Depressed": "#E07B7B"},
            ax=axes[1], linewidth=1.5)
axes[1].set_title("Age Boxplot by Depression Status", fontweight="bold")
axes[1].set_xlabel("")
axes[1].set_ylabel("Age")

plt.tight_layout()
plt.savefig("viz2_age_distribution.png", bbox_inches="tight")
plt.show()
```



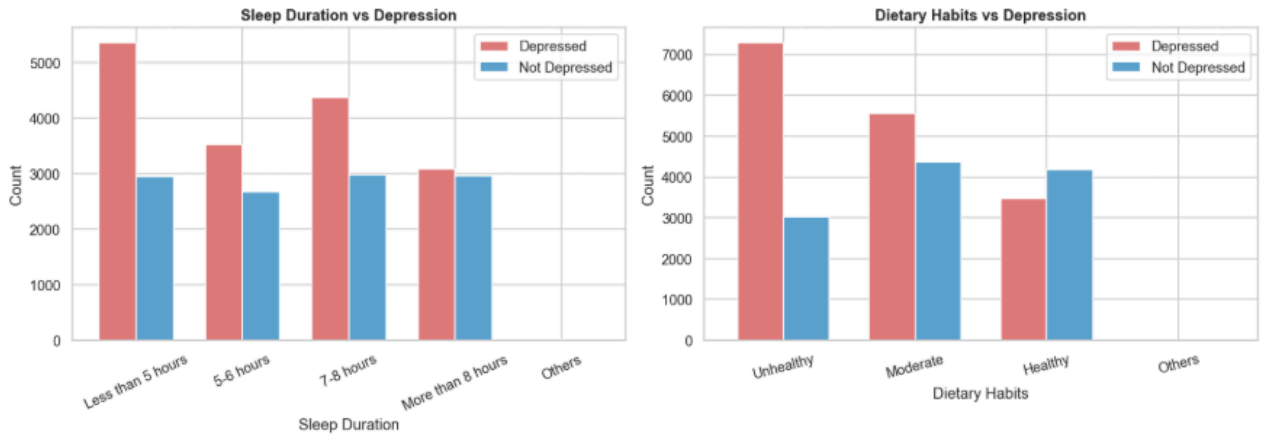
### 5.3 Academic Pressure and CGPA

```
fig, axes = plt.subplots(1, 2, figsize=(13, 5))

# --- Left: How depression changes with academic pressure level ---
sns.countplot(data=df, x="Academic Pressure", hue="Depression_Label",
              palette={"Not Depressed": "#5BA4CF", "Depressed": "#E07B7B"},
              ax=axes[0], edgecolor="white")
axes[0].set_title("Academic Pressure vs Depression", fontweight="bold")
axes[0].set_xlabel("Academic Pressure (1=Low, 5=High)")
axes[0].set_ylabel("Count")
axes[0].legend(title="Status")

# --- Right: CGPA distribution for depressed vs not ---
for val, label, color in [(0, "Not Depressed", "#5BA4CF"), (1, "Depressed", "#E07B7B")]:
    subset = df[df["Depression"] == val]["CGPA"]
    axes[1].hist(subset, bins=25, alpha=0.65, label=label, color=color, edgecolor="white")

axes[1].set_title("CGPA Distribution by Depression Status", fontweight="bold")
axes[1].set_xlabel("CGPA")
axes[1].set_ylabel("Frequency")
axes[1].legend()

plt.tight_layout()
plt.savefig("viz3_academic.png", bbox_inches="tight")
plt.show()
```



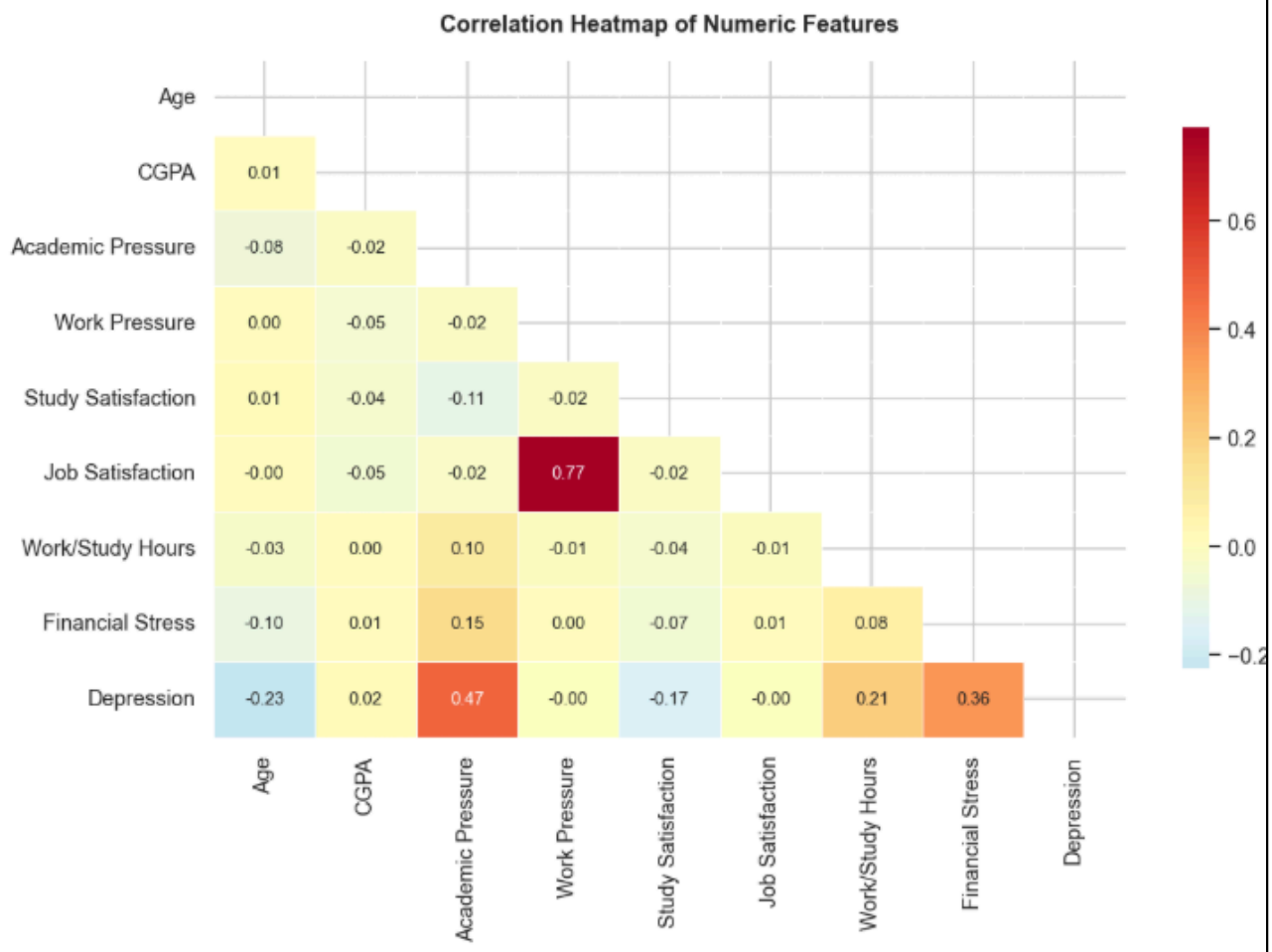
## 5.4 Sleep Duration and Dietary Habits

```
# Define the order we want to display these categories
sleep_order = ["Less than 5 hours", "5-6 hours", "7-8 hours", "More than 8 hours", "Others"]
diet_order  = ["Unhealthy", "Moderate", "Healthy", "Others"]

fig, axes = plt.subplots(1, 2, figsize=(14, 5))

# --- Left: Sleep duration vs depression ---
# Group and count, then reshape for easy plotting
sleep_data = df.groupby(["Sleep Duration", "Depression_Label"]).size().unstack(fill_value=0)
sleep_data = sleep_data.reindex([s for s in sleep_order if s in sleep_data.index])
sleep_data.plot(kind="bar", ax=axes[0], color=["#E07B7B", "#5BA4CF"], edgecolor="white", width=0.7)
axes[0].set_title("Sleep Duration vs Depression", fontweight="bold")
axes[0].set_xlabel("Sleep Duration")
axes[0].set_ylabel("Count")
axes[0].tick_params(axis="x", rotation=25)
axes[0].legend()

# --- Right: Dietary habits vs depression ---
diet_data = df.groupby(["Dietary Habits", "Depression_Label"]).size().unstack(fill_value=0)
diet_data = diet_data.reindex([d for d in diet_order if d in diet_data.index])
diet_data.plot(kind="bar", ax=axes[1], color=["#E07B7B", "#5BA4CF"], edgecolor="white", width=0.7)
axes[1].set_title("Dietary Habits vs Depression", fontweight="bold")
axes[1].set_xlabel("Dietary Habits")
axes[1].set_ylabel("Count")
axes[1].tick_params(axis="x", rotation=15)
axes[1].legend()

plt.tight_layout()
plt.savefig("viz4_sleep_diet.png", bbox_inches="tight")
plt.show()
```



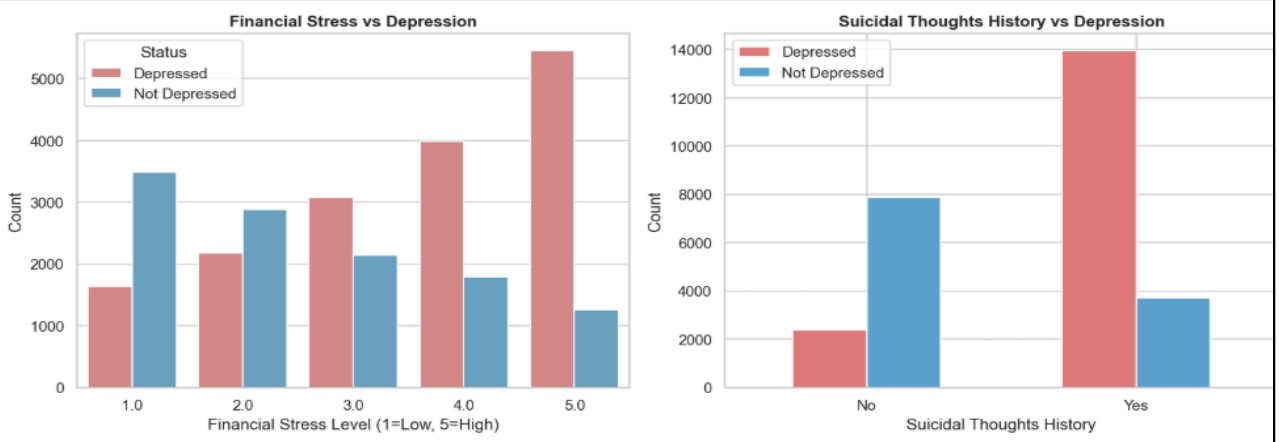
### 5.5 Correlation Heatmap of Numeric Features

```
# Pick only the number columns for correlation
num_cols = ["Age", "CGPA", "Academic Pressure", "Work Pressure",
            "Study Satisfaction", "Job Satisfaction",
            "Work/Study Hours", "Financial Stress", "Depression"]

# Calculate how related each pair of columns is (-1 to 1)
corr = df[num_cols].corr()

# Only show the bottom half of the table (the top half is the same)
mask = np.triu(np.ones_like(corr, dtype=bool))

fig, ax = plt.subplots(figsize=(10, 7))
sns.heatmap(corr, mask=mask, annot=True, fmt=".2f",
            cmap="RdYlBu_r", center=0, linewidths=0.5,
            ax=ax, annot_kws={"size": 9}, cbar_kws={"shrink": 0.8})
ax.set_title("Correlation Heatmap of Numeric Features", fontweight="bold", pad=15)
plt.tight_layout()
plt.savefig("viz5_heatmap.png", bbox_inches="tight")
plt.show()
```



### 5.6 Financial Stress and Suicidal Thoughts

```
fig, axes = plt.subplots(1, 2, figsize=(13, 5))

# --- Left: Financial stress level vs depression ---
sns.countplot(data=df, x="Financial Stress", hue="Depression_Label",
              palette={"Not Depressed": "#5BA4CF", "Depressed": "#E07B7B"},
              ax=axes[0], edgecolor="white")
axes[0].set_title("Financial Stress vs Depression", fontweight="bold")
axes[0].set_xlabel("Financial Stress Level (1=Low, 5=High)")
axes[0].set_ylabel("Count")
axes[0].legend(title="Status")

# --- Right: Suicidal thoughts history vs depression ---
suicidal_data = df.groupby(["Have you ever had suicidal thoughts ?", "Depression_Label"]).size().unstack(fill_value=0)
suicidal_data.plot(kind="bar", ax=axes[1], color=["#E07B7B", "#5BA4CF"], edgecolor="white", width=0.55)
axes[1].set_title("Suicidal Thoughts History vs Depression", fontweight="bold")
axes[1].set_xlabel("Suicidal Thoughts History")
axes[1].set_ylabel("Count")
axes[1].tick_params(axis="x", rotation=0)
axes[1].legend()

plt.tight_layout()
plt.savefig("viz6_stress_suicidal.png", bbox_inches="tight")
plt.show()
```




## 6. Observations and Interpretation

After looking at the data, here are five things I noticed:

**1. More students are depressed than not.** Out of 27,901 students in my dataset, about 58.5% are depressed and 41.5% are not. This means the data is a little unbalanced, with more depressed students than non-depressed ones. This is something to keep in mind when building a prediction model later. I might also trim the data a bit so that it will be balanced.

**2. Higher academic pressure is linked to more depression.** Students who rated their academic pressure as 4 or 5 (out of 5) were more likely to be depressed. This makes sense because the more stressed a student is about school, the harder it is to stay mentally healthy.

**3. Students who sleep less tend to be more depressed.** Students who sleep less than 5 hours a night made up the biggest group of depressed students. On the other hand, students who sleep 7 to 8 hours tend to have better mental health. Getting enough sleep seems to really matter.

**4. Eating habits and money problems also play a role.** Students with unhealthy diets had higher depression rates compared to those who eat well. Also, students with higher financial stress (level 4 or 5) were more often depressed. This shows that everyday problems like money and food choices can affect mental health too.

**5. Academic pressure and financial stress are the strongest warning signs.** Looking at the heatmap, academic pressure and financial stress had the highest connection to depression among all the numeric columns. Interestingly, CGPA (grades) had a very weak connection, meaning getting good or bad grades alone does not really predict if a student is depressed.